

Organizing Scientific Knowledge from the Engineering Sciences Using Research Knowledge Graphs

Dear participant,

We cordially invite you to participate in our survey to evaluate the ongoing work to organize scientific knowledge in engineering using research knowledge graphs. Based on a specific use case on **Tailored Forming Process Chains**, we started to collect, extract, and analyze scientific knowledge from publications. In this context, we have organized various topics to describe process chains in the engineering sciences in a structured manner and thus capture this knowledge systemically.

Your subject matter expertise and feedback are invaluable in assessing the usefulness of the topics and their organization.

The survey is divided into three parts:

1. **Demographics:** Here we ask you to provide some demographic information.
2. **Evaluation of the topics:** Here we ask you to rate the usefulness of the topics for engineering sciences.
3. **Evaluation of the organization of the topics:** Here we ask you to indicate your agreement with the structuring of the topics for engineering sciences.

Your answers will be treated confidentially and used exclusively for research purposes.

Participation in the survey takes about **10-15 minutes**.

We thank you in advance for your time and valuable contributions.

Yours sincerely,

Oliver Karras (TIB - Leibniz Information Centre for Science and Technology),
Laura Budde (Laser Zentrum Hannover e.V.),
and Paulina Merkel (IPH - Institut für Integrierte Produktion Hannover gGmbH)

* Gibt eine erforderliche Frage an

Demographics

1. In which subject area of engineering sciences are you a professional? *

We are using the DFG subject area classification. If you need further details to decide if your discipline belongs to one of the subject areas, refer to the following [DFG publication](#).

Engineering Sciences				
Mechanical and Industrial Engineering	Thermal Engineering/ Process Engineering	Materials Science and Engineering	Computer Science, Systems and Electrical Engineering	Construction Engineering and Architecture

Markieren Sie nur ein Oval.

- ☐ Mechanical and Industrial Engineering
- ☐ Thermal Engineering / Process Engineering
- ☐ Material Sciences and Engineering
- ☐ Computer Science, Systems, and Electrical Engineering
- ☐ Construction Engineering and Architecture
- ☐ None

2. How many years of professional experience do you have in engineering sciences? *

3. What is the highest educational degree you have obtained in engineering sciences? *

Markieren Sie nur ein Oval.

- ☐ Bachelor's Degree
- ☐ Master's Degree
- ☐ Doctorate (Ph.D.)
- ☐ Sonstiges: _____

Evaluation of the Topics

Below, we provide definitions of the **topics** for the description of process chains:

- **Materials:** The materials used in a process chain to manufacture a component.
- **Process chains:** The sequences of process steps for manufacturing a component.
- **Process steps:** The steps of a process for manufacturing a component.
- **Inputs:** The inputs, e.g., material or intermediate component, of a process step.
- **Outputs:** The outputs, e.g., intermediate or final component, of a process step.
- **Manufacturing methods:** The manufacturing methods used in a process step to manufacture the output of the process step.
- **Manufactured components:** The output, e.g., intermediate or final component, resulting from a process step.
- **Layers:** The material layers of a manufactured component.
- **Qualities:** The qualities of a manufactured component measured in a process step.
- **Measurement methods:** The measurement methods used to measure the qualities of a manufactured component in a process step
- **Measurements:** The measurements, i.e., values and units, of the qualities of a manufactured component in a process step that result from the measurements method used.

4. For each topic listed below, please assess the following statement: The topic ... is **useful** for describing process chains in engineering sciences. *

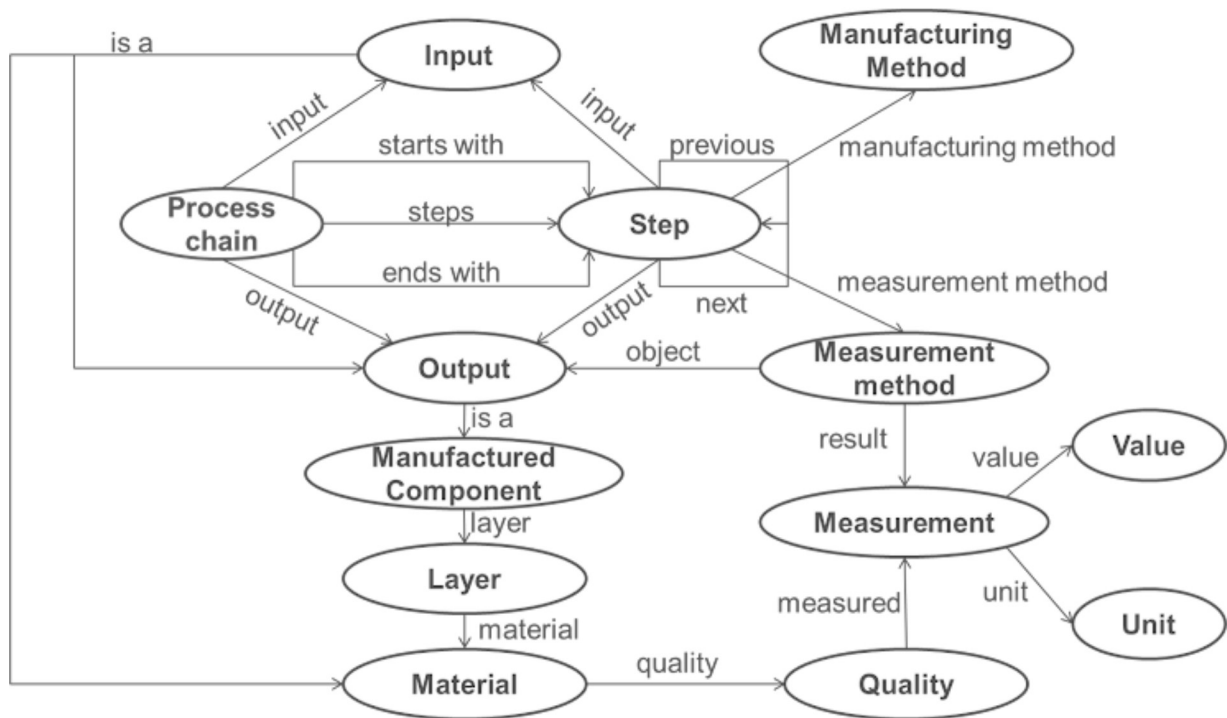
Markieren Sie nur ein Oval pro Zeile.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Materials	☐	☐	☐	☐	☐
Process chain	☐	☐	☐	☐	☐
Process Steps	☐	☐	☐	☐	☐
Inputs	☐	☐	☐	☐	☐
Outputs	☐	☐	☐	☐	☐
Manufacturing methods	☐	☐	☐	☐	☐
Manufactured components	☐	☐	☐	☐	☐
Layers	☐	☐	☐	☐	☐
Qualities	☐	☐	☐	☐	☐
Measurement methods	☐	☐	☐	☐	☐
Measurements	☐	☐	☐	☐	☐

Evaluation of the Organization of the Topics

Below, we provide an overview of the structuring of the **topics** for the description of process chains.

Structuring of the topics



We have formulated each part of the figure as a statement of our modeling.

Please indicate how much you agree with each of the following ten statements.

5. A **Material** has one or more qualities that are measured. *

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

6. A **Process chain** has two or more process steps, a start process step, an end ^{*} process step, an input, and an output.

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly agree

7. A **Process Step** can have a previous process step and a next process step, ^{*} and it has an input, an output, one or more manufacturing methods, and one or more measurement methods.

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

8. An **Input** is either one or more materials or the output of the previous process ^{*} step.

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

9. An **Output** is a manufactured component. ^{*}

Markieren Sie nur ein Oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

10. A **Manufacturing method** is the method used in a process step to manufacture the output of the process step. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

11. A **Manufactured component** has one or more layers. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

12. A **Layer** consists of a material. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

13. A **Quality** is measured by one or more measurements. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

14. A **Measurement method** has an output of a process step as the object of investigation and one or more measurements. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

15. A **Measurement** has a value and a unit. *

Markieren Sie nur ein Oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

Thank You For Your Participation

We appreciate you taking the time to complete our survey. Your insights and expertise are invaluable to us and will greatly contribute to the success of our research.

If you have any additional comments or suggestions, please feel free to share them by email. We are always looking for ways to improve and value your feedback.

Help Us Reach More Experts!

If you know other professionals from engineering sciences who might be interested in participating, please share this survey with them. Your support in spreading the word is greatly appreciated!

If you have any questions or need further assistance, please contact us at oliver.karras@tib.eu.

Thank you once again for your valuable contribution!

Best regards,

Oliver Karras (TIB - Leibniz Information Centre for Science and Technology),
Laura Budde (Laser Zentrum Hannover e.V.),
and Paulina Merkel (IPH - Institut für Integrierte Produktion Hannover gGmbH)

Dieser Inhalt wurde nicht von Google erstellt und wird von Google auch nicht unterstützt.

Google Formulare